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SASexp3a.m | SASexp4a.m

```
1   % Experiment 11: Linearity and Time Invariance
2 - clc; clear; close all;
3 - t = -1:0.001:1;
4 - x1 = sin(2*pi*5*t); x2 = cos(2*pi*5*t);
5 - a = 2; b = 3; t0 = 0.2;
6 - T = @(x) 2*x;
7 - lhs = T(a*x1+b*x2);
8 - rhs = a*T(x1)+b*T(x2);
9 - linError = max(abs(lhs-rhs));
10 - xDelay = sin(2*pi*5*(t-t0));
11 - lhsTI = T(xDelay);
12 - rhsTI = T(x1);
13 - rhsTIshift = interp1(t,rhsTI,t-t0,'linear',0);
14 - tiError = max(abs(lhsTI-rhsTIshift));
15 - figure;
16 - subplot(2,1,1); plot(t,lhs-rhs); grid on; title('Linearity Error');
17 - subplot(2,1,2); plot(t,lhsTI-rhsTIshift); grid on; title('Time-Invariance Error');
18 - fprintf('Linearity error = %g\n',linError);
19 - fprintf('Time-invariance error = %g\n',tiError);
```

Workspace

Name	Value
a	2
b	3
lhs	*1x2001 double*

Command Window

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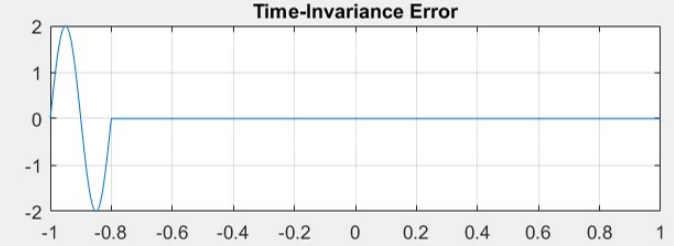
SASexp3a.m | SASexp4a.m

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Figure 1

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Workspace

Name	Value
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b	3
lhs	*1x2001 double*

Ln 19 Col 1

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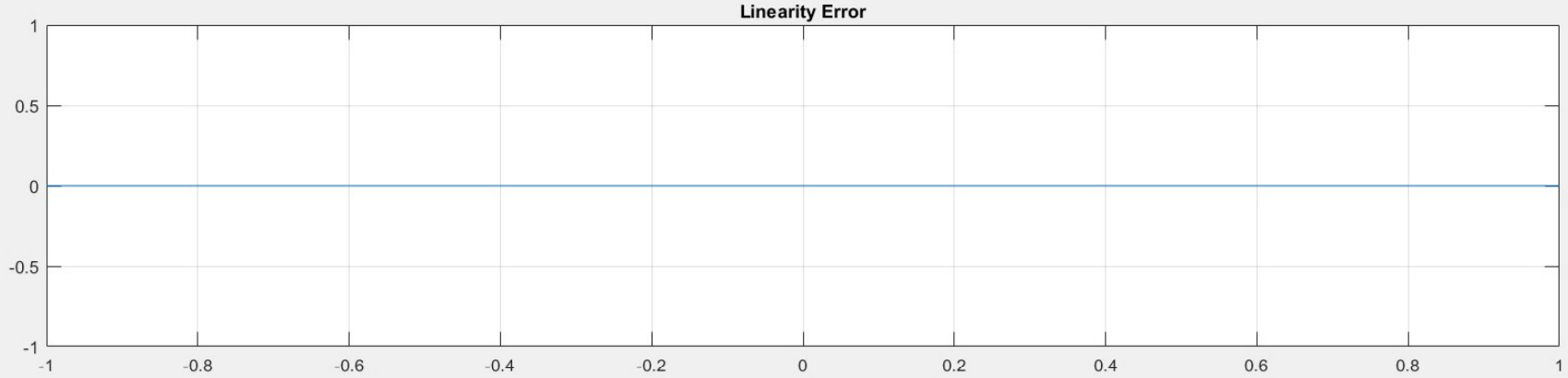
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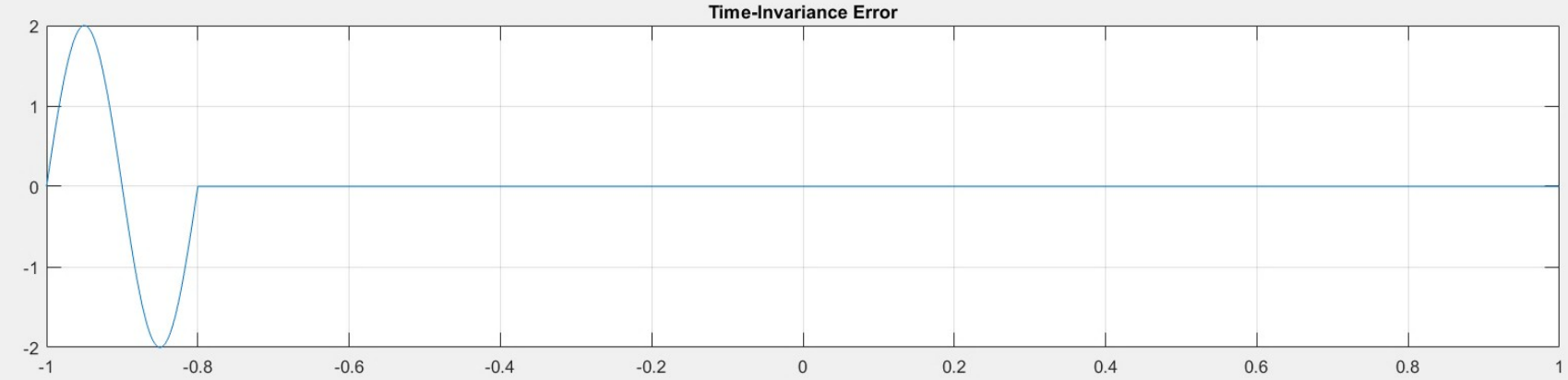
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Figure 1

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